

FinBench

When to trust AI with financial figures: a calibration benchmark built on SEC filings.

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Five leading AI models were each asked 358 questions whose answers sit in SEC filings, three times each, with no documents and no web access: the way an analyst types a question into a chat window. The main measure is how often a model gives a **wrong figure while saying it is at least 75% sure**. That is the answer most likely to be copied into a model or a memo unchecked.

20.8%

of the budget model's answers were wrong and stated with high confidence

0.1–5.5%

for the four flagship models on the same questions

0

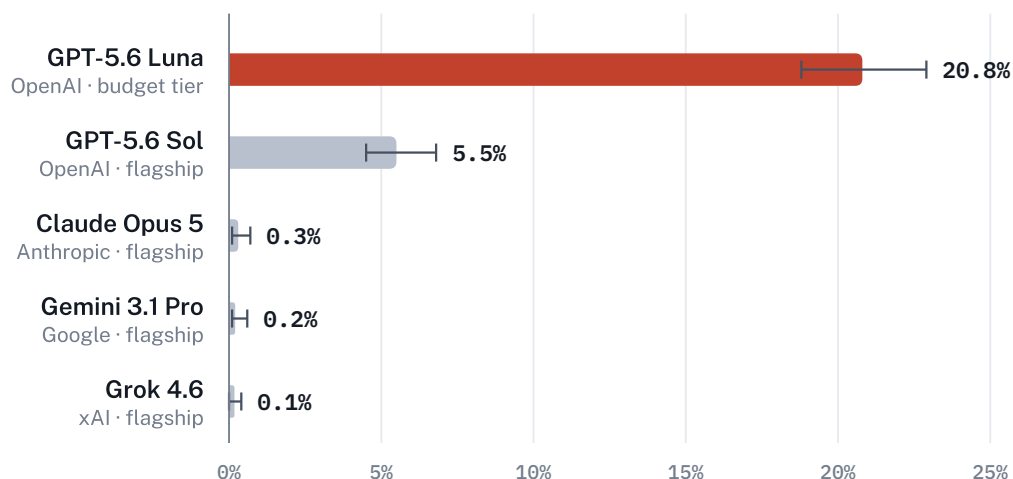
figures invented across 570 answers about business segments that don't exist

Confident-wrong: a figure the filing contradicts, given with stated confidence of 75 or more. Accuracy is graded at the precision the model stated.

The budget model is confidently wrong about one time in five

GPT-5.6 Luna, OpenAI's low-cost model, gave a confident wrong figure on 223 of its 1,074 answers. The flagship from the same generation, GPT-5.6 Sol, did so on 59. The other three flagships each did so fewer than four times.

EXHIBIT 1 Share of answers that were confidently wrong

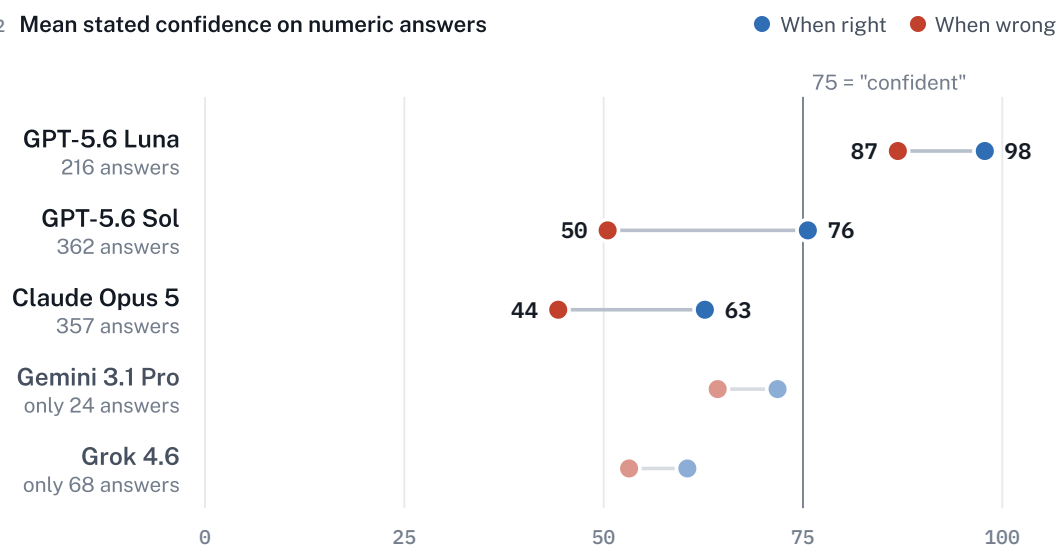


Bars show the rate; lines show the 90% Wilson interval. n = 1,074 answers per model (358 questions × 3). Source: FinBench run d849113d369c; ground truth from SEC 10-K and 10-Q filings.

The gap is in the confidence, not the knowledge

None of these models reliably recalls exact figures from filings. What separates them is what they say when they don't know. When Luna gave a wrong number it still reported a mean confidence of 87, barely below the 98 it reported when right. Sol drops to 50 when wrong, and Claude Opus 5 to 44. Their stated confidence carries a warning; Luna's does not.

EXHIBIT 2 Mean stated confidence on numeric answers



Confidence is the 0–100 figure each model reported alongside its answer. Gemini and Grok declined most numeric questions, so their averages rest on few answers and are shown faded. Source: FinBench run d849113d369c.

Mistakes cluster where a plausible old number exists

Every closed-book model handled the traps: made-up segments, obscure figures and periods after its training data all drew zero confident errors. The errors come from two kinds of question. One is a **restated figure**, where the model remembers the number as originally reported. The other is a period **just before its training cutoff**, where it fills the gap with a guess.

EXHIBIT 3 Confidently wrong, by type of question					
	GPT-5.6 Luna	GPT-5.6 Sol	Claude Opus 5	Gemini 3.1 Pro	Grok 4.6
Restated figures a number the company later revised	41%	4%	0	0.9%	0.4%
Just before the cutoff a period near the model's training cutoff	44%	17%	1%	0	0
Obscure segment figures real but rarely quoted numbers	0	0	0	0	0
After the cutoff periods the model cannot have seen	0	0	0	0	0
Segments that don't exist a false premise the model must reject	0	0	0	0	0

Share of each model's answers in that category, darker = higher. n per model: restated 231, near cutoff 294, obscure 291, after cutoff 144, non-existent segment 114. Source: FinBench run d849113d369c.

What this means for an analyst

- Check any figure for a period near the model's training cutoff, whatever confidence it states. That is where confident errors peak, reaching 44% of answers for the budget model.
- Expect restated numbers to come back as originally reported. After a spin-off or an accounting change, a model will often give the old figure and be sure of it.
- Read the confidence field only on models that use it. A flagship's stated confidence falls sharply when it is wrong. The budget model's barely moves, so a cheaper model saves money at the cost of that warning.

How the results were checked

Answers come from the filings

358 questions built from XBRL data in 10-K and 10-Q filings across 50 companies, chosen for messy structures (spin-offs, 52-week fiscal years, share classes) rather than size.

Rules fixed before any model ran

Grading rules were committed before the first model call, and the git history shows the order. The rules were never changed after results came in.

Answer key verified by hand

A seeded sample of 78 answer-key records was checked against the primary filings: 78 of 78 confirmed. An automated review beforehand found 13 questions asking for figures that companies later revised; that question type was reworded before the final run.

Results reproduce

260 unchanged questions were run a week apart, and no model's results moved beyond noise ($|z| \leq 1.4$). The same check flagged a web-search reference model whose provider changed its behaviour under an unchanged name, so it is reported separately and not compared here.

Models were called through OpenRouter with provider fallbacks disabled and any substituted model quarantined; 3 samples per question; all 6,444 answers graded. Rates carry 90% Wilson intervals. Limitations: 280 of the 358 answer-key records are verified by the automated pipeline only, per-category samples are 114–294 answers per model, and this is the closed-book track only, with no documents provided.

[Full technical report](#) [Methodology](#) [Code and data](#)